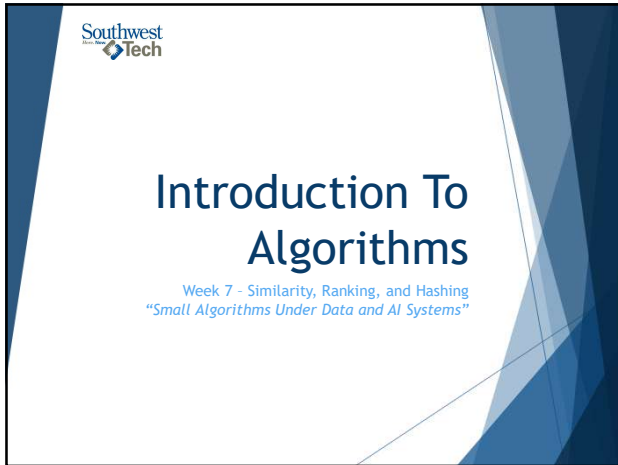
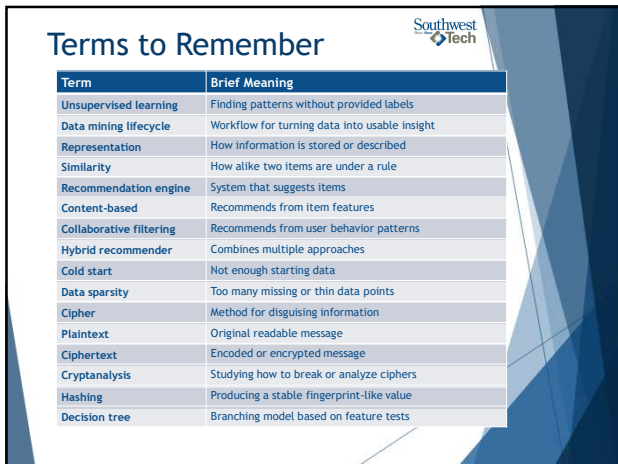




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3

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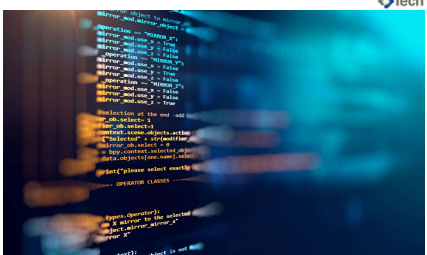
What We Will “Touch On” and Revisit Later

- ▶ Full machine-learning workflows
- ▶ Larger datasets
- ▶ Model training
- ▶ Collaborative filtering at scale
- ▶ Cryptography beyond simple ciphers and hashing
- ▶ Decision trees and supervised learning in more depth
- ▶ Security, privacy, and responsible AI/data practice



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Review Coding Lab

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Review: What Lab 06 Taught Us

Last week - Representation shaped the result

- What the nodes represented
- What the edges represented
- Where traversal started
- What the model included
- What the model left out

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Today's Question

How can a small algorithm compare, rank, recommend, group, or verify data?

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Small Algorithms Under Larger Systems

Larger systems often rely on small algorithmic choices:

- ▶ represent the data
- ▶ compare items
- ▶ score matches
- ▶ rank results
- ▶ group similar items
- ▶ check identity or integrity



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Algorithms in Systems

AI / Data System

1. Input Data

Item	Feature A	Feature B	Feature C
Item 1	8	7	6
Item 2	5	9	7
Item 3	7	6	8
Item 4	6	5	6
Item 5	4	6	5

2. Scoring Rule

compare features

3. Ranking Output

Item	Score
Item 2	8.7
Item 3	7.4
Item 1	7.0
Item 4	5.7
Item 5	4.6

4. Larger System

Used by the larger system

- Make decisions
- Personalize
- Recommend
- Improve over time

Small, clear algorithmic steps work together inside the system to produce useful results.

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What Counts as Success Today

- make one AI/data idea visible
- explain the representation
- show evidence
- name assumptions
- name a limitation
- avoid overstating the result

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Textbook Review

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Four Reading Areas

- 1. unsupervised learning
- 2. recommendation engines
- 3. cryptography basics
- 4. decision trees as recommended context

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Textbook Review

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Unsupervised Learning

Unsupervised learning looks for patterns without provided answer labels.

Examples:

- ▶ grouping similar items
- ▶ finding clusters
- ▶ detecting structure
- ▶ discovering patterns

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Algorithm
Text + Text
Text + Graph
Text + Table

ALGORITHMS
Design and Analysis

DATA STRUCTURES
& ALGORITHMS

DISCRETE MATHEMATICS

COMPUTER SCIENCE
FUNDAMENTALS

PROBLEM SOLVING
WITH ALGORITHMS

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Data Mining Lifecycle

1. business understanding
2. data understanding
3. data preparation
4. modeling
5. evaluation
6. deployment



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Algorithm
Text + Text
Text + Graph
Text + Table

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Recommendation Engines

Recommendation engines suggest options by using data.

Common types:

- ▶ content-based
- ▶ collaborative filtering
- ▶ hybrid

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Algorithm

Textbook Review

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Cryptography Basics

The cryptography reading introduces:

- ▶ weakest link
- ▶ security goals
- ▶ sensitive data
- ▶ ciphers
- ▶ plaintext and ciphertext
- ▶ cryptanalysis

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Algorithm

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Decision Trees

Recommended reading introduces decision trees.

For this week:

- ▶ recognize the branching idea
- ▶ notice strengths and weaknesses
- ▶ save deeper use for later AI/data courses

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Design and Analysis

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& ALGORITHMS

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Deeply Learn

Deeply learn now:

- ▶ representation
- ▶ similarity
- ▶ ranking
- ▶ assumptions
- ▶ limitations
- ▶ visible evidence

Recognize for later:

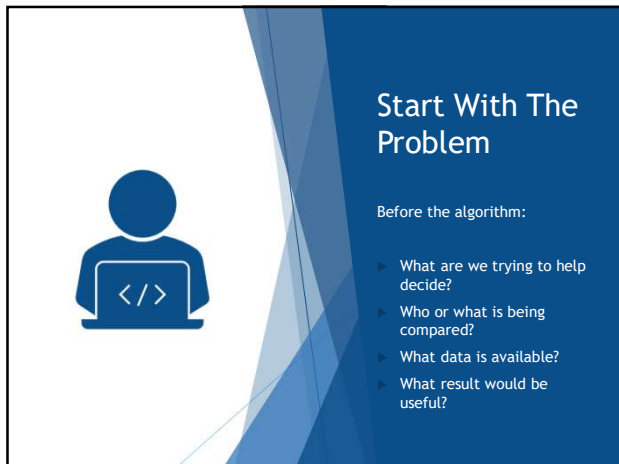
- ▶ full ML workflows
- ▶ advanced recommendation systems
- ▶ full cryptography
- ▶ decision-tree modeling



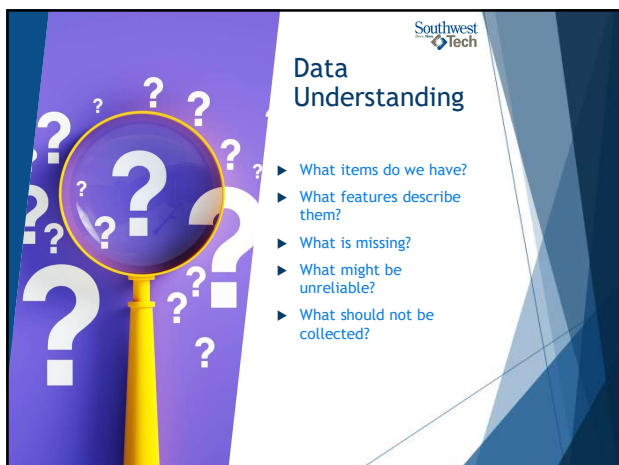
18



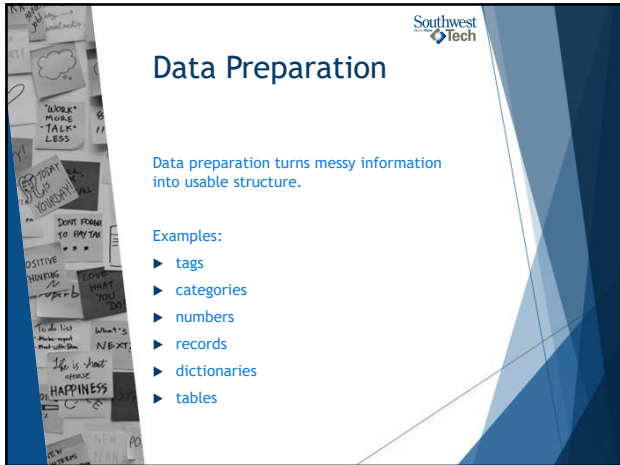
19



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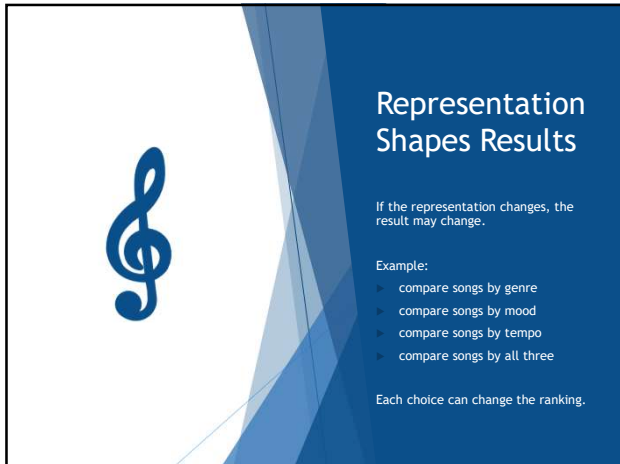
Data Preparation

Data preparation turns messy information into usable structure.

Examples:

- tags
- categories
- numbers
- records
- dictionaries
- tables

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Representation Shapes Results

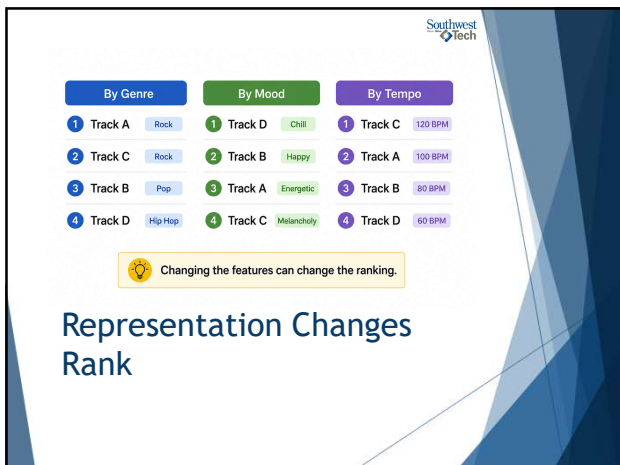
If the representation changes, the result may change.

Example:

- compare songs by genre
- compare songs by mood
- compare songs by tempo
- compare songs by all three

Each choice can change the ranking.

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Representation Changes Rank

By Genre		By Mood		By Tempo	
1 Track A	Rock	1 Track D	Chill	1 Track C	120 BPM
2 Track C	Rock	2 Track B	Happy	2 Track A	100 BPM
3 Track B	Pop	3 Track A	Energetic	3 Track B	80 BPM
4 Track D	Hip Hop	4 Track C	Melancholy	4 Track D	60 BPM

Changing the features can change the ranking.

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Similarity

Similarity is a rule for deciding how alike two things are.

It may use:

- ▶ shared tags
- ▶ matching categories
- ▶ numeric distance
- ▶ text overlap
- ▶ feature weights

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Scores Need Meaning

A score should answer:

- ▶ What does a high score mean?
- ▶ What does a low score mean?
- ▶ Are all features equal?
- ▶ What score range is possible?
- ▶ What does the score ignore?

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Similarity, Ranking, and Recommendation



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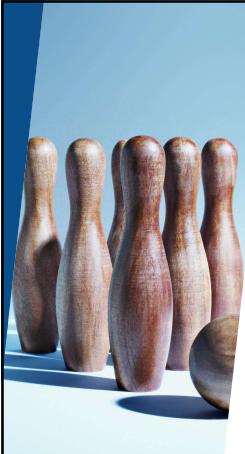


Ranking

Ranking orders items by a score or rule.

Ranking can show:

- ▶ strongest match
- ▶ weakest match
- ▶ top recommendation
- ▶ items needing review



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Recommended Resources

	Resource	Score
1	Practice Set B	92
2	Video Walkthrough A	87
3	Reference Sheet C	74
4	Challenge Task D	61

Scoring Rule (Transparent)

 +
  +
 

topic match + difficulty fit + recent progress

Score = topic match + difficulty fit + recent progress

 The top item is highest by this rule, not universally best.

Ranking

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Content-Based Recommendation

Content-based recommendation uses item features.

Example:

- ▶ "If a user likes items with tags A and B, recommend other items with similar tags."

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Similarity In Unstructured Documents

Text can be compared by features such as:

- ▶ words
- ▶ tags
- ▶ topics
- ▶ categories
- ▶ extracted terms

The representation still matters.

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Collaborative Filtering



Collaborative filtering uses patterns from users or behavior.

It may ask:

- ▶ What did similar users choose?
- ▶ What did people with similar histories like?
- ▶ What patterns appear across many users?

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Collaborative Filtering Risks



Risks include:

- ▶ limited sample size
- ▶ over-reliance on history
- ▶ isolated analysis
- ▶ reinforcing past patterns
- ▶ weak recommendations for new users

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Hybrid Recommendation

Hybrid recommendation combines signals.

It may use:

- ▶ item features
- ▶ user history
- ▶ similarity scores
- ▶ reference vectors
- ▶ feedback over time

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Similarity Matrix And Reference Vector

Two useful structures:

- ▶ similarity **matrix**: compares items to items
- ▶ reference **vector**: represents a user, target, or profile

Both make comparison visible.

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	Item A	Item B	Item C	Item D
Item A	1.0	0.8	0.3	0.6
Item B	0.8	1.0	0.4	0.7
Item C	0.3	0.4	1.0	0.5
Item D	0.6	0.7	0.5	1.0

→

compare patterns

Reference Vector

Genre match 0.8

Mood match 0.6

Tempo match 0.4

Difficulty fit 0.7

💡 Each item's pattern is compared to the user's reference vector.

Similarity Matrix And Reference Vector

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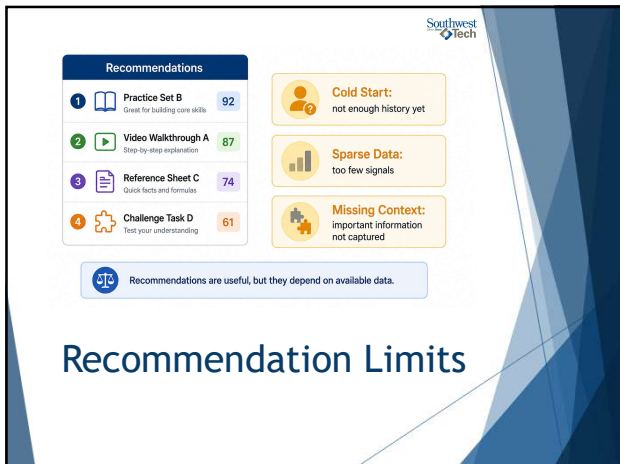
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Recommendation Limits

Recommendation systems have limits:

- ▶ cold start
- ▶ metadata requirements
- ▶ data sparsity
- ▶ social influence
- ▶ missing context
- ▶ biased or incomplete data

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Recommendation Limits

Recommendations

- 1 Practice Set B 92
Great for building core skills
- 2 Video Walkthrough A 87
Step-by-step explanation
- 3 Reference Sheet C 74
Quick facts and formulas
- 4 Challenge Task D 61
Test your understanding

Cold Start:
not enough history yet

Sparse Data:
too few signals

Missing Context:
important information not captured

Recommendations are useful, but they depend on available data.

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Cold Start And Sparse Data

Cold start:

- ▶ not enough information at the beginning

Data sparsity:

- ▶ too many blanks or thin signals

Both weaken recommendations.

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Social Influence Is Double-Edged

Social influence can:

- ▶ reveal useful patterns
- ▶ amplify popularity
- ▶ hide better-fit options
- ▶ reinforce existing bias
- ▶ make recommendations feel objective

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Clustering As Grouping By Closeness

Clustering groups items by closeness or similarity.

(For Lab 07), Clustering can be simple:

- ▶ choose features
- ▶ compare items
- ▶ create groups
- ▶ explain why the groups make sense
- ▶ name what the grouping does not prove

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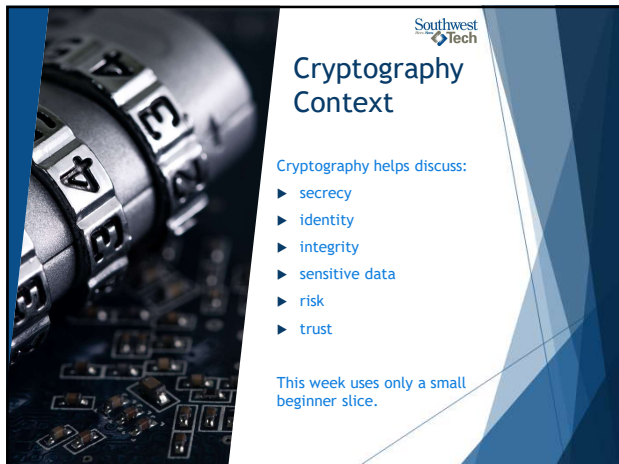
Break

10 Minute "Human" bio-break

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Security Requirement Frame

Before choosing a method:

1. identify the entities
2. establish security goals
3. understand data sensitivity

The slide features a background with a blue fingerprint graphic and a circuit board pattern.

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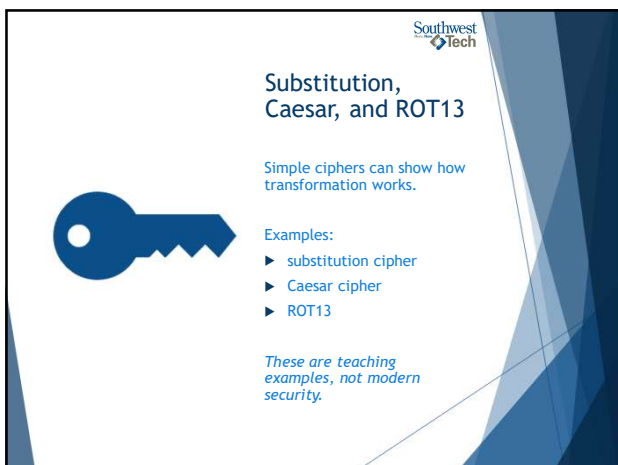
Cryptography Terms

Terms to recognize:

- ▶ cipher
- ▶ plaintext
- ▶ ciphertext
- ▶ cipher suite
- ▶ encryption
- ▶ decryption
- ▶ cryptanalysis
- ▶ PII

The slide features a background with a keyboard and a blue geometric design.

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Substitution, Caesar, and ROT13

Simple ciphers can show how transformation works.

Examples:

- ▶ substitution cipher
- ▶ Caesar cipher
- ▶ ROT13

These are teaching examples, not modern security.

The slide features a large blue key icon on the left and a blue geometric design on the right.

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Cryptanalysis And Transposition

 Cryptanalysis asks how a method might be analyzed or broken.

 Transposition ciphers rearrange positions rather than substituting symbols.

 Both show that algorithm design includes attack thinking.

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Hashing



Hashing produces a stable value from input data.

Useful for:

- ▶ identity checks
- ▶ integrity checks
- ▶ lookup support
- ▶ change detection

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Hashing as Integrity Checking

	Text Record	Hash Value
Original Record	Lab score = 92	A7F3
Changed Record	Lab score = 93	9C21

 Small input change, different hash.

 Hashing checks change; it does not hide the original content.

Hashing

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Hashing vs Encryption

Hashing:

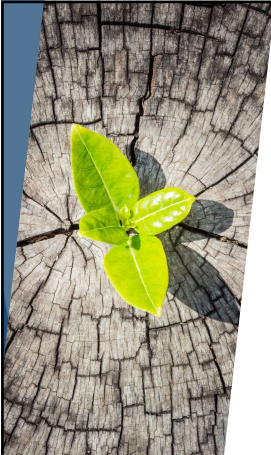
- ▶ one-way check value
- ▶ useful for identity or integrity

Encryption:

- ▶ protects readable content
- ▶ should be decryptable by authorized parties

Do not treat them as the same.

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Decision Tree Recognition

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Decision Tree Recognition

A decision tree uses branching questions.

Example shape:

- ▶ Is the feature present?
- ▶ Is the value above a threshold?
- ▶ Which branch applies?
- ▶ What class or decision results?

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Decision Tree Strengths And Weaknesses

Strengths

- ▶ explainable structure
- ▶ visible decision path
- ▶ useful for classification



Weaknesses

- ▶ can overfit
- ▶ depends on selected features
- ▶ can become brittle



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Demo Scenario

Problem:

Goal

Create a tiny music recommendation example.

- Represent songs with simple tags such as genre, mood, tempo, and instrument.
- Compare one reference song to several candidate songs using a basic similarity score based on shared tags.

Use:

- a reference song
- candidate songs
- simple tags
- tag-overlap similarity
- ranked recommendations

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Demo Evidence

The demo produces:

- ▶ item-feature table
- ▶ reference song summary
- ▶ similarity score table
- ▶ ranked recommendation
- ▶ assumption and limitation notes

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Demo Key Questions

- ▶ Which tags are being compared?
- ▶ What does the score mean?
- ▶ Which feature is missing?
- ▶ Why did the top item win?
- ▶ What should the result not claim?

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Presentation Layer Progression

Same logic, different presentation:

1. plain tables
2. ANSI color
3. Rich tables and panels

The algorithm should not change.

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Console Presentation Styles

Plain

Scan started...
File: data.csv
Size: 2.1 MB
Status: PASS
Issues found: 0
File: report.txt
Size: 18 KB
Status: MAIN
Issues found: 2
File: notes.log
Size: 124 KB
Status: FOUND
Issues found: 5
Scan complete.

ANSI Color

Scan started...
File: data.csv
Size: 2.1 MB
Status: PASS
Issues found: 0
File: report.txt
Size: 18 KB
Status: MAIN
Issues found: 2
File: notes.log
Size: 124 KB
Status: FOUND
Issues found: 5
Scan complete.

Rich

File	Size	Status	Issues
data.csv	2.1 MB	PASS	0
report.txt	18 KB	MAIN	2
notes.log	124 KB	FOUND	5

Scan complete.

💡 Presentation layer can improve readability without changing the algorithm.

Presentation Styles

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Demo Explanation

"The top recommendation ranked highest because it shared the most tags with the reference song. This result depends on the tags I chose, and it does not prove the listener would actually prefer the song."

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From Demo to Lab

In Lab 07, choose or receive one option:

1. similarity ranking
2. simple clustering
3. tiny recommendation
4. hashing demonstration

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Lab 07 Evidence

similarity matrix

ranking table

cluster grouping table

scatter plot

hash comparison

representation table

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Four Acceptable Forms of Lab Evidence

1 Ranking Table

Item	Score
1. Practice Set B	92
2. Video Walkthrough A	87
3. Reference Sheet C	74
4. Challenge Task D	61

2 Cluster Table

Cluster	Items
Cluster 1	Item A, Item B, Item C
Cluster 2	Item D, Item E, Item F

3 Hash Comparison

Record	Hash Value
Original Record	A7F3
Changed Record	9C21

4 Representation Table

Item	Genre	Mood	Tempo	Difficulty
Item A	Rock	Energetic	Fast	Medium
Item B	Pop	Happy	Medium	Easy
Item C	Jazz	Calm	Slow	Hard
Item D	Hip Hop	Focused	Medium	Medium

Evidence should make the reasoning visible.

Evidence Choices

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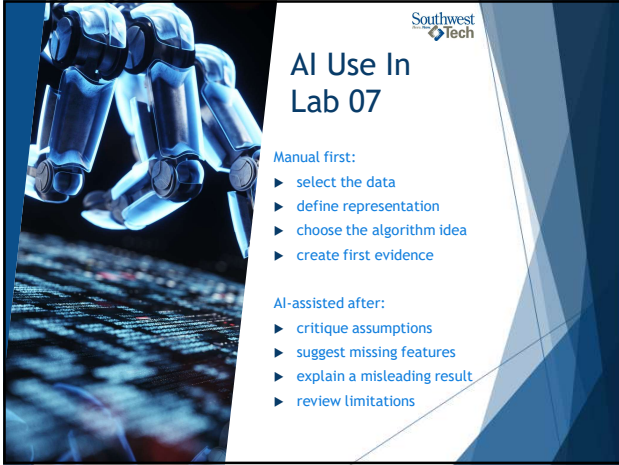
README Evidence



Your README should show:

- ▶ selected option
- ▶ data set
- ▶ representation explanation
- ▶ visible evidence
- ▶ two assumptions
- ▶ one limitation or risk
- ▶ AI/data connection
- ▶ AI-use note, if applicable

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AI Use In Lab 07

Manual first:

- ▶ select the data
- ▶ define representation
- ▶ choose the algorithm idea
- ▶ create first evidence

AI-assisted after:

- ▶ critique assumptions
- ▶ suggest missing features
- ▶ explain a misleading result
- ▶ review limitations

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Wrap up

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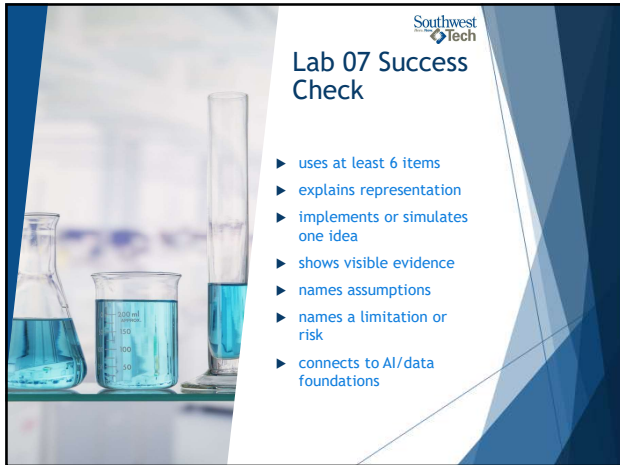
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What To Carry Forward

Carry forward:

- ▶ representation shapes results
- ▶ similarity is a design choice
- ▶ rankings need explanation
- ▶ small data has limits
- ▶ hashing can support integrity
- ▶ evidence must be visible

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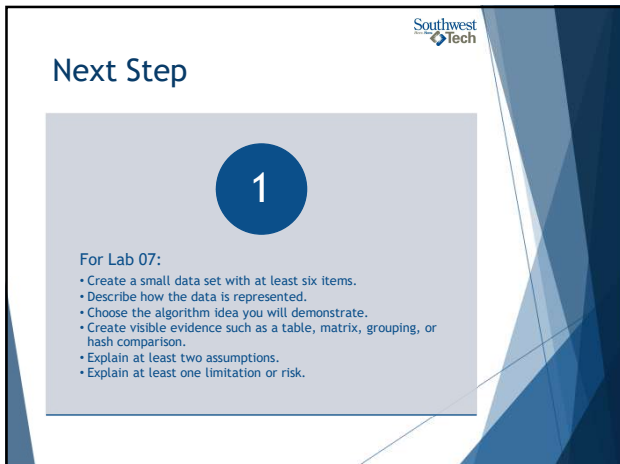


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Lab 07 Success Check

- ▶ uses at least 6 items
- ▶ explains representation
- ▶ implements or simulates one idea
- ▶ shows visible evidence
- ▶ names assumptions
- ▶ names a limitation or risk
- ▶ connects to AI/data foundations

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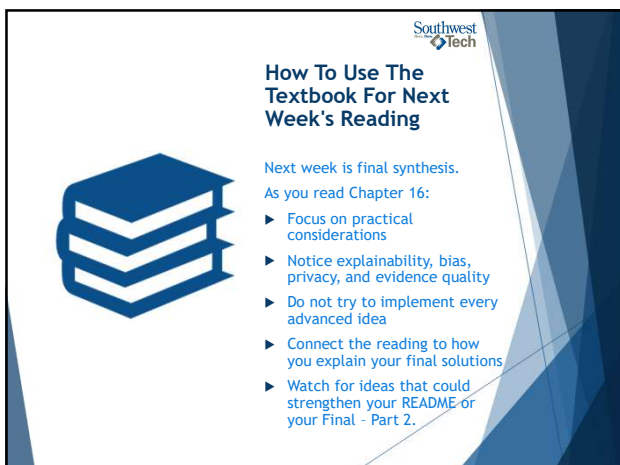
Next Step

1

For Lab 07:

- Create a small data set with at least six items.
- Describe how the data is represented.
- Choose the algorithm idea you will demonstrate.
- Create visible evidence such as a table, matrix, grouping, or hash comparison.
- Explain at least two assumptions.
- Explain at least one limitation or risk.

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How To Use The Textbook For Next Week's Reading



Next week is final synthesis.
As you read Chapter 16:

- ▶ Focus on practical considerations
- ▶ Notice explainability, bias, privacy, and evidence quality
- ▶ Do not try to implement every advanced idea
- ▶ Connect the reading to how you explain your final solutions
- ▶ Watch for ideas that could strengthen your README or your Final - Part 2.

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